METRIC
TT-C-492D(SH)
May 15, 2024
SUPERSEDING
TT-C-492C
May 31, 1974

FEDERAL SPECIFICATION

COATINGS, ANTI-SWEAT

The General Services Administration has authorized the use of this federal specification by all federal agencies, some states and localities.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers coatings applied by spray, with brush application for touch-up or repair, to provide anti-sweat protection on surfaces subject to condensation.

1.2 Classification. Coatings shall be of the following types, classes, grades, and applications, as specified (see 6.2).

1.2.1 Types.

a. Type I – Type I coatings shall have a volatile organic content (VOC) content not greater than 50 grams per liter (0.4 pounds per gallon).

b. Type II – Type II coatings shall have a VOC content not greater than 150 grams per liter (1.3 pounds per gallon).

c. Type III – Type III coatings shall have a VOC content not greater than 250 grams per liter (2.1 pounds per gallon).

1.2.2 Classes.

a. Class 1 – Class 1 coatings shall have a thermal conductivity less than or equal to 0.08 watts/(meter × kelvin) [0.05 British thermal units/(hour × foot × °F)] or [0.56 British thermal units × inch/(hour × square foot × °F)].

b. Class 2 – Class 2 coatings shall have a thermal conductivity greater than 0.08 watts/(meter × kelvin) ([0.05 British thermal units/(hour × foot × °F)] or [0.56 British thermal units × inch/(hour × square foot × °F)]).

c. Class 3 – Class 3 coatings shall prevent condensation by water absorption or water vapor absorption.

1.2.3 Grades.

a. Grade A – Grade A coatings shall prevent condensation (see 3.4.1) for not less than 24 hours.

b. Grade B – Grade B coatings shall prevent condensation (see 3.4.1) for not less than 8 hours.

1.2.4 Applications.

a. Application A – Application A coatings shall have a maximum final dry film thickness (DFT) (see 3.3.e) greater than 3.2 millimeters (⅛ inch).

b. Application B – Application B coatings shall have a maximum final DFT (see 3.3.e) equal to or not greater than 3.2 millimeters (⅛ inch).

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the accuracy of the address information using the ASSIST Online database at https://assist.dla.mil.

FSC 8010

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FEDERAL SPECIFICATIONS

TT-P-645 - Paint, Alkyd Type, Formula Number 84

(Copies of this document are available online at https://quicksearch.dla.mil/.)

FEDERAL STANDARDS

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing

(Copies of this document are available online at https://quicksearch.dla.mil/.)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-23236 - Coating Systems for Ship Structures

MIL-DTL-24441 - Paint, Epoxy-Polyamide, General Specification for

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

FEDERAL REGULATIONS

40 CFR 60, Appendix A-7, Method 24 - Determination of Volatile Matter Content, Water Content, Density, Volume Solids, and Weight Solids of Surface Coatings

40 CFR 63, Subpart II - National Emission Standards for Shipbuilding and Ship Repair (Surface Coating)

40 CFR 63, Appendix A, Method 311 - Analysis of Hazardous Air Pollutant Compounds in Paints and Coatings by Direct Injection Into a Gas Chromatograph

40 CFR 82 - Protection of Stratospheric Ozone

40 CFR 261.24 - Toxicity Characteristic

(Copies of these documents are available online at www.ecfr.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AK-DPC-010/078-1 - Composite Materials, Surface Ships, Topside Structural and Other Topside Applications – Fire Performance Requirements

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of these documents are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

ASSOCIATION FOR MATERIALS PROTECTION AND PERFORMANCE (AMPP)

SSPC-SP 1	-	Solvent Cleaning
SSPC-SP 10 (WAB)/ NACE No. 2 (WAB)	-	Near-White Metal Wet Abrasive Blast Cleaning
SSPC-SP 11	-	Power Tool Cleaning to Bare Metal
SSPC-SP 17	-	Thorough Abrasive Blast Cleaning of Non-Ferrous Metals

(Copies of these documents are available online at www.ampp.org.)

ASTM INTERNATIONAL

ASTM A1011/1011M	-	Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength
ASTM C518	-	Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus
ASTM D1640/D1640M	-	Standard Test Methods for Drying, Curing, or Film Formation of Organic Coatings
ASTM D1849	-	Standard Test Method for Package Stability of Paint
ASTM D3273	-	Standard Test Method for Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber
ASTM D3278	-	Standard Test Methods for Flash Point of Liquids by Small Scale Closed-Cup Apparatus
ASTM D4400	-	Standard Test Method for Sag Resistance of Paints Using a Multinotch Applicator
ASTM D6450	-	Standard Test Method for Flash Point by Continuously Closed Cup (CCCFP) Tester
ASTM D6677	-	Standard Test Method for Evaluating Adhesion by Knife
ASTM E104	-	Standard Practice for Maintaining Constant Relative Humidity by Means of Aqueous Solutions
ASTM E662	-	Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.

ASTM E1252 - Standard Practice for General Techniques for Obtaining Infrared Spectra for Qualitative Analysis

ASTM F718 - Standard Specification for Shipbuilders and Marine Paints and Coatings Product/Procedure Data Sheet

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL MARITIME ORGANIZATION (IMO)

Fire Test Procedures (FTP) Code

(Copies of this document are available online at www.imo.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org/home.html.)

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

NIOSH Manual of Analytical Methods

(Copies of this document are available online at www.cdc.gov/niosh.)

3. REQUIREMENTS

3.1 Composition. The coatings shall comply with all occupational health and safety regulations and shall meet all requirements specified herein.

3.1.1 Hazardous pigments and additives. When tested as specified in 4.5.2.1, the metal content (soluble and total) of the coating shall be less than the values listed in table I. If the test for total metal results in a value less than the soluble metal limit, the soluble metal test need not be conducted. Crystalline silica shall not exceed 0.1 percent by weight (wt%).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.

TABLE I. Metals content.

Metal and its compound in each coating dry film	Soluble metal, maximum (milligrams per liter)	Total metal content, maximum wt%
Antimony	15.0	0.015
Arsenic	5.0	0.005
Barium (excluding barite)	100.0	0.10
Beryllium	0.75	0.0002
Cadmium	1.0	0.0005
Chromium (VI)	1.0	0.001
Chromium (including chromium III)	560.0	0.56
Cobalt	50.0	0.005
Copper	25.0	0.01
Fluoride salts	180.0	0.18
Lead	5.0	0.005
Mercury	0.2	0.0002
Molybdenum	350.0	0.035
Nickel	20.0	0.02
Selenium	1.0	0.002
Silver	5.0	0.001
Tantalum	100.0	0.100
Thallium	7.0	0.007
Tungsten	100.0	0.100
Vanadium	24.0	0.01
Zinc	250.0	0.25

3.1.2 Toxicity. When evaluated in accordance with 4.5.2.2, the coating shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.5.2.2 and 6.4).

3.1.3 Prohibited materials. The coating shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS411-1 for all other programs (see 4.5.2.3).

3.1.4 Hazardous air pollutants (HAPs) content. When evaluated in accordance with 4.5.2.4, the content of the HAPs solvents in the coating shall be less than the wt% values listed in table II. The definition of HAPs materials are in accordance with 40 CFR 63, subpart II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.

TABLE II. Hazardous solvents.

Hazardous solvent in mixed coating or its components	Maximum, wt%
Benzene	0.05
Chlorinated solvent(s), total	0.05
Solvents containing fluorine in accordance with 40 CFR 82	0.01
Ethyl benzene	0.05
Methyl-, ethyl-, and butyl-mono-ethers of ethylene glycol or the acetates thereof, total (also known as methyl-, ethyl-, and butyl-cello solves acetates)	0.05
Methyl ethyl ketone	0.05
Methyl isobutyl ketone	0.05
Toluene	0.05
Xylene (all forms), total	0.1

3.2 First article inspection. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.3 Identification (ID) characteristics. Values for ID characteristics shall be those established for the product at the time the product is submitted for conformance inspection. The purpose of these values is to serve as a basis for determining that the coating and individual coating components being offered are essentially the same as that which was approved under conformance inspection. Unless otherwise specified herein, ID characteristics for all types, classes, grade, and applications shall be conducted at 16 to 27 °C (60 to 80 °F) and below 85 percent relative humidity. The manufacturer shall provide ASTM methods, other consensus standards, or a copy of unique test methodology or any other necessary information on methodology used to determine reported ID characteristics (see 3.3.a through m).

a. Color
b. Unit of issue
c. Percent volatiles
d. Pot life for two-component systems
e. Maximum final DFT for the coating system
f. Density for all liquid coating components
g. Chemical nature of each system component (e.g., epoxy, acrylic, waterborne acrylic, latex, etc.)
h. Application method(s)
i. Mix ratios of each system component/mixing instructions for two-component systems
j. Shelf life
k. Required storage and transportation temperature range
l. Maximum wet film thickness (WFT) per coat
m. Minimum final dry film thickness (mDFT) for condensation protection

3.4 Quantitative requirements. The coating shall meet the quantitative requirements specified in 3.4.1 through 3.4.9.2.3 and table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.

TABLE III. Quantitative requirements.

Quantitative property	Classification	Standard	Requirements	Test methods	Conformance testing
Condensation	All	----	3.4.1	4.5.3	No
Dry time	All	----	3.4.2	4.5.4	Yes
Sag resistance	All	ASTM D4400	3.4.3	4.5.5	Yes
Knife adhesion	All	ASTM D6677	3.4.4	4.5.6	No
Thermal conductivity	Class 1 Class 2	ASTM C518	3.4.5, Table IV	4.5.7	No
Flash point	All	ASTM D3278 ASTM D6450	3.4.6	4.5.8	Yes
VOC	All	40 CFR 60, appendix A-7, method 24	3.4.7	4.5.9	Yes
Mold/mildew resistance	All	ASTM D3273	3.4.8	4.5.10	No
Fire performance	All	----	3.4.9	4.5.11	No
Surface flammability	Application A	IMO Fire Test Procedures (FTP) Code, annex 1, part 5	3.4.9.1.1	4.5.11.1.1	No
Smoke density	Application A	ASTM E662	3.4.9.1.2	4.5.11.1.2	No
Fire gas toxicity	Application A	T9070-AK-DPC-010/078-1	3.4.9.1.3	4.5.11.1.3	No
Critical radiant flux	Application B	IMO Fire Test Procedures (FTP) Code, annex 1, part 5	3.4.9.2.1	4.5.11.2.1	No
Smoke density	Application B	ASTM E662	3.4.9.2.2	4.5.11.2.2	No
Fire gas toxicity	Application B	T9070-AK-DPC-010/078-1	3.4.9.2.3	4.5.11.2.3	No

3.4.1 Condensation. When tested in accordance with 4.5.3, there shall be no visible condensation on the drip pan for the 24-hour test period for grade A coatings or the 8-hour test period for grade B coating. The test coating shall not exhibit any loss of adhesion, sagging, corrosion, running rust, blistering, wrinkling, or peeling during the 24-hour test period for grade A coatings or the 8-hour test period for grade B coatings.

3.4.2 Dry time. When tested in accordance with 4.5.4, dry-through time for the coating shall be not greater than 24 hours.

3.4.3 Sag resistance. When tested in accordance with 4.5.5, the coating system shall not sag when applied at a WFT of 60 mils.

3.4.4 Knife adhesion. When tested in accordance with 4.5.6, the adhesion between the coating and the primer shall have a rating not less than 4. When recoated, the adhesion between the coating and previous layer of coating shall have a rating not less than 4.

3.4.5 Thermal conductivity. When tested in accordance with 4.5.7, the thermal conductivity measured for class 1 and class 2 coatings shall meet the requirements of table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.

TABLE IV. Thermal conductivity values for each class.

Class	Thermal conductivity [watts per (meter × kelvin)]	Thermal conductivity [British thermal units/(hour × foot × °F)]	Thermal conductivity [British thermal units × inch/(hour × square foot × °F)]
1	Less than or equal to 0.08	Less than or equal to 0.05	Less than or equal to 0.56
2	Greater than 0.08	Greater than 0.05	Greater than 0.56
3	Not applicable	Not applicable	Not applicable

3.4.6 Flash point. When tested in accordance with 4.5.8, the flash point of the coating or any component shall be not less than 38 °C (100 °F).

3.4.7 VOC. When tested in accordance with 4.5.9, the VOC of a coating shall not exceed the maximum content prescribed for the specified type (see 1.2.1).

3.4.8 Mold/mildew resistance. When tested in accordance with 4.5.10, the test panels shall have a rating of 8 or greater.

3.4.9 Fire performance. When tested in accordance with 4.5.11, the fire performance shall meet the requirements of 3.4.9.1 for application A and 3.4.9.2 for application B.

3.4.9.1 Application A fire performance. When tested in accordance with 4.5.11.1, the fire performance of application A coatings shall meet the requirements of 3.4.9.1.1 to 3.4.9.1.3.

3.4.9.1.1 Surface flammability. When tested in accordance with 4.5.11.1.1, the coating shall meet the following surface flammability requirements:

a. The critical flux at extinguishment (CFE) shall be greater than or equal to 20 kilowatts per square meter (6,340 British thermal units per hour per square foot).

b. The heat for sustained burning (Qsb) shall be greater than or equal to 1.5 megajoules per square meter (132 British thermal units per square foot).

c. The total heat release (Qt) shall be less than or equal to 0.7 megajoules (663.5 British thermal units).

d. The peak heat release rate (Qp) shall be less than or equal to 4 kilowatts (13,650 British thermal units per hour).

e. No melting, dripping, or flaming droplets shall be evident.

3.4.9.1.2 Smoke density. When tested in accordance with 4.5.11.1.2, the application A coating shall have a smoke density not greater than 200.

3.4.9.1.3 Fire gas toxicity. When tested in accordance with 4.5.11.1.3, the test results shall meet the fire gas toxicity requirements in accordance with T9070-AK-DPC-010/078-1.

3.4.9.2 Application B fire performance. When tested in accordance with 4.5.11 and 4.5.11.2, the fire performance of application B coatings shall meet the requirements of 3.4.9.2.1 to 3.4.9.2.3.

3.4.9.2.1 Critical radiant flux. When tested in accordance with 4.5.11.2.1, the application B coating shall have a critical radiant flux not less than 0.45 watt per square centimeter.

3.4.9.2.2 Smoke density. When tested in accordance with 4.5.11.2.2, the application B coating shall have a smoke density not greater than 200.

3.4.9.2.3 Fire gas toxicity. When tested in accordance with 4.5.11.2.3, the test results shall meet the fire gas toxicity requirements in accordance with T9070-AK-DPC-010/078-1.

3.5 Qualitative requirements. The coating shall meet the qualitative requirements specified in table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.

TABLE V. Qualitative requirements.

Qualitative property	Classification	Standard	Requirements	Test methods	Conformance testing
Color	All	----	3.5.1	4.6.1	Yes
Water resistance	All	----	3.5.2	4.6.2	No
Compatibility with primers and undercoats	All	----	3.5.3	4.6.3	No
Accelerated storage stability	All	ASTM D1849 FED-STD-141, method 3011.3	3.5.4	4.6.4	No
Spraying properties	All	FED-STD-141, method 4331.2	3.5.5	4.6.5	No

3.5.1 Color. When tested in accordance with 4.6.1, colors of each individual layer of the coating system shall visually match the color identified by the manufacturer of each layer (see 3.3.a).

3.5.2 Water resistance. When tested in accordance with 4.6.2, the coating shall not blister, wrinkle, peel or delaminate when soaked in potable water for 1 hour.

3.5.3 Compatibility with primers and undercoats. When tested in accordance with 4.6.3, the coating shall show no signs of incompatibility, such as blistering, wrinkling, peeling, or delamination from the primer or the undercoats.

3.5.4 Accelerated storage stability. When tested in accordance with 4.6.4, a freshly opened full container of the coating shall be free of grit, seeds, skins, lumps, putrefaction, rancidity, or livering. Any pigment settling, or caking or material separation shall be easily mixed back into a smooth homogeneous state. Surface float (separated pigments or other ingredients on the surface of the liquid coating) will occur in many anti-sweat coatings and mixing in accordance with the manufacturer's commercial product data sheet or the NAVSEA-reviewed ASTM F718 product data sheet shall readily reincorporate the float resulting in a smooth, homogenous state. The container shall be free of leaks, ill-fitting lids, and corrosion.

3.5.5 Spraying properties. When tested in accordance with 4.6.5, the coating shall show no running, sagging, or streaking. The air-dried film shall show no seeding, dusting, floating, mottling, hazing, or other film defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his/her own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of examinations and tests. Examinations and tests shall be classified as follows:

a. First article inspection (4.3).

b. Quality conformance inspection (4.4).

4.3 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on a random sample drawn from the first production lot or batch. First article inspection shall include all tests specified herein.

4.3.1 Changes to product. A change in material, production processes, or production equipment used in the manufacture of the coatings shall require the coatings to be retested in accordance with requirements herein or waived via written approval by the Technical Authority (see 6.6). Incorporation of any changes that have not been approved shall require retesting of the coating.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection testing shall consist of the tests specified in table III and table V. At a minimum, quality conformance inspections specified in table III and table V shall be performed on the first lot and every 19,000 liters (5,000 gallons) thereafter. A lot shall consist of an individual system component of the same type, class, grade, and application from a single uniform batch or a uniform blend of batches offered for delivery at one time.

4.5 Test methods. The coatings shall be tested in accordance with the applicable methods specified herein.

4.5.1 Specimens and conditions.

4.5.1.1 Panel preparation. Test plates of the nature and size specified in the applicable test method shall be coated in accordance to the manufacturer's application instructions or the NAVSEA-reviewed ASTM F718 product data sheet with a primer conforming to TT-P-645 or a primer qualified to either MIL-DTL-24441, type IV or MIL-PRF-23236, type V, VI, or VII. Unless otherwise specified herein, test panels shall be cold-rolled steel plates from steel conforming to ASTM A1011/1011M or as approved by the Technical Authority (minimum size: 102 by 152 by 3 millimeters [4 by 6 by ⅛ inches] [nominal]). Each individual coat of the overall coating system shall be mixed and applied in accordance with the manufacturer's commercial product data sheet or the NAVSEA-reviewed ASTM F718 product data sheet. Prior to priming, the test panels shall be prepared in accordance with SSPC-SP 10 (WAB)/NACE No. 2 (WAB), SSPC-SP 11, or SSPC-SP 17.

4.5.1.2 Controlled laboratory conditions. Unless otherwise specified herein, all preparation, application, and testing shall be conducted under controlled laboratory conditions. Controlled laboratory conditions shall be maintained between 16 to 27 °C (60 to 80 °F) and below 85 percent relative humidity.

4.5.2 Composition.

4.5.2.1 Hazardous pigments and additives. Soluble and total metal content shall be determined in accordance with 40 CFR 261.24. Calculation of individual hazardous metal contents can be based on either the manufacturer's testing of lots or batches in accordance with 40 CFR 261.24 or the supplier's data for raw materials used in the product. Soluble metal content shall be recorded as milligrams per liter. Crystalline silica shall be analyzed in accordance with the NIOSH Manual of Analytical Methods, method 7500 with the coating sample treated as a bulk settled dust. Test results shall be in accordance with the requirements of 3.1.1.

4.5.2.2 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.1.2 as specified (see 6.2). The Navy and Marine Corps Force Health Protection Command (NMCFHPC) will evaluate the coating system using data provided by the manufacturer/distributor to the NMCFHPC (see 3.1.2 and 6.4).

4.5.2.3 Prohibited materials. Prohibited materials shall be verified as specified (see 3.1.3 and 6.2).

4.5.2.4 HAPs content. The HAPs content of the coating shall be measured in accordance with 40 CFR 63, appendix A, method 311 (Environmental Protection Agency test method 311). Solvent fractions shall be identified in accordance with ASTM E1252 with the results recorded as wt% of the total coating system. Alternate methods of analysis shall be approved by Technical Authority. Formulation data may be used by manufacturers in lieu of testing to demonstrate compliance with 3.1.4. The formulation data shall have a consistent and quantitatively known relationship to the testing required. Calculation of individual HAPs content can be based on either the manufacturer's evaluation of batches or supplier data for raw materials used in the product. The test results for HAPs content shall meet the requirements of 3.1.4.

4.5.2.5 ID characteristics. Unless otherwise specified (see 6.2), the coating shall be visually inspected and identification characteristics shall be reviewed to ensure conformance to the requirements specified in 3.3.

4.5.3 Condensation. The condensation test setup shall be composed of the condensation enclosure and a drip pan as shown in figure 1. The condensation enclosure shall be hung above the drip pan inside a chamber of sufficient size to hold one or more of the condensation enclosures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.

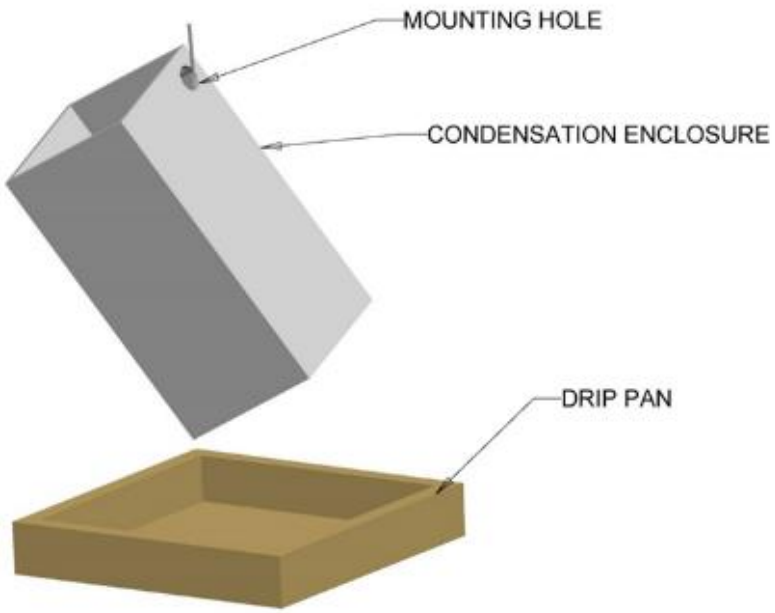
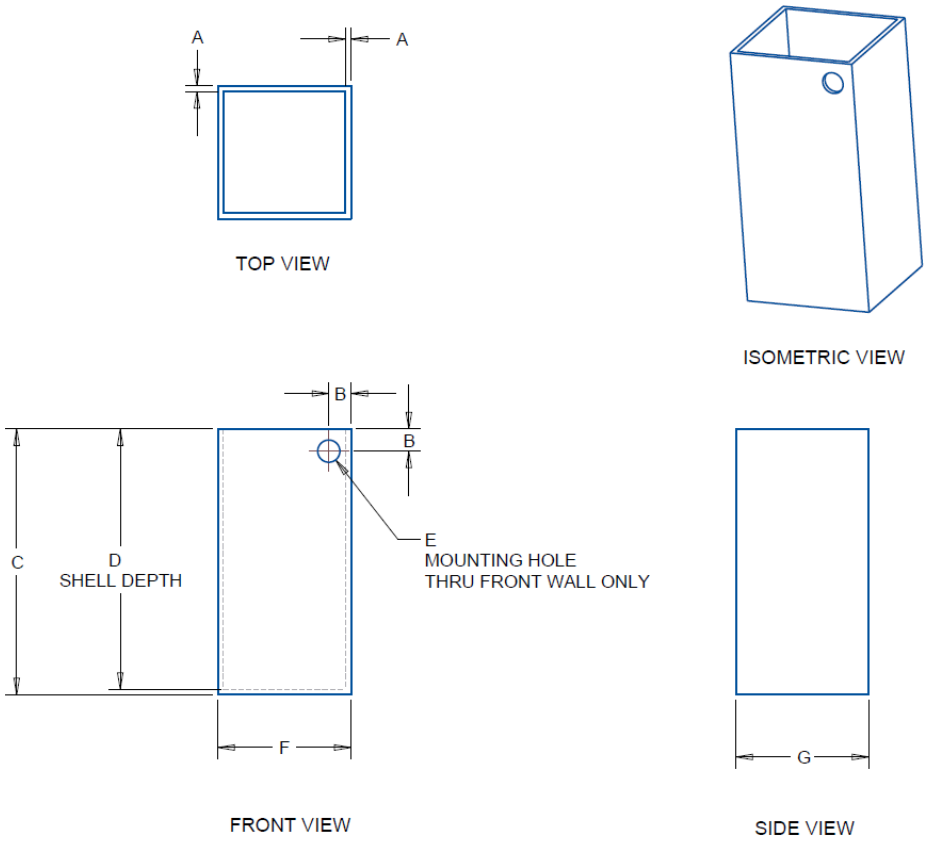


FIGURE 1. Condensation test setup.

4.5.3.1 Condensation enclosure. The condensation enclosure shall be composed of steel, aluminum, or copper. The condensation enclosure shall meet the designs and dimensions in figure 2. The condensation enclosure shall be not less than 30 centimeters (12 inches) long. The condensation enclosure shall have a maximum wall thickness of 6.5 millimeters (0.25 inch) and should be not less than 0.5 millimeter (0.02 inch). The bottom plate of the condensation enclosure shall be of the same thickness as the wall of the condensation enclosure. The bottom plate shall be welded, adhesively bonded, soldered, or otherwise attached to form a flat, watertight end. All welds shall be flush with each surface. No weld splatter shall be present on the surface of the condensation enclosure. One side of the condensation enclosure shall have a mounting hole with a maximum diameter of 13 millimeters (0.5 inch) nominal as shown in figure 1 and figure 2 with the midpoint of the mounting hole residing 2.5 centimeters (1 inch) from the top and nearest side.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.



DIMENSION NOTES:

A. Maximum of 6.5 millimeters (0.25 inch).

B. Maximum of 13 millimeters (0.5 inch).

C. Minimum length of 30.5 centimeters (12 inches).

D. Not less than 30 centimeters (11.75 inches).

E. Not greater than 13 millimeters (0.5 inch).

F. Not less than 10 centimeters (4 inches) and not greater than 15.25 centimeters (6 inches).

G. Not less than 10 centimeters (4 inches) and not greater than 15.25 centimeters (6 inches).

FIGURE 2. Condensation test enclosure schematic.

4.5.3.2 Coating preparation and application. All sides, bottom, welds, and corners of the condensation enclosure shall be hand sanded or orbital sanded with 80- to 120-grit aluminum oxide paper to create a uniform random pattern of scratches on the surface. Prior to coating, the condensation enclosure shall be solvent cleaned and be in accordance with the cleanliness requirements of SSPC-SP 1. The exterior of the condensation enclosure shall be coated with a primer conforming to TT-P-645 or a primer qualified to either MIL-DTL-24441, type IV or MIL-PRF-23236, type V, VI, or VII in accordance with the manufacturer's commercial product data sheet or the NAVSEA-reviewed ASTM F718 product data sheet. The anti-sweat coating shall be applied to the primed surface at the mDFT (see 3.3.m) in accordance with the manufacturer's commercial product data sheet. The coating shall be allowed to dry as directed by the manufacturer's commercial product data sheet before the condensation enclosure is ready for use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.

4.5.3.3 Temperature and humidity control. The temperature within the condensation enclosure shall be maintained through use of a potable water-ice mixture or through circulating a chilled liquid solution through the condensation enclosure at a temperature between 0 and 3.3 °C (32 and 38 °F). When the condensation enclosure is placed on a flat surface such that the mounting hole is oriented as shown in figure 2, Front View, the enclosure shall be filled with the potable water-ice mixture or chilled liquid solution until the condensation enclosure is nominally between ½ and ¾ full. During the test, the temperature of the potable water-ice mixture or other liquid solution shall be maintained at a temperature between 0 and 3.3 °C (32 and 38 °F). The potable water-ice mixture or other liquid solution shall be maintained, circulated, or replaced as needed to maintain the specified temperature. The condensation enclosure may include a removable, insulating top fitted to the open end of the enclosure to reduce the rate at which the ice melts. Ice shall be added as needed to maintain the required fluid temperature inside of the condensation enclosure. Liquid may be removed as needed to ensure that the volume of liquid shall not exceed ¾ of the volume of the condensation enclosure. Adding ice or removing liquid may result in mechanical jarring of the condensation enclosure that may result in condensation dripping into the drip pan. Condensation dripping off of the condensation enclosure during the replenishment of ice or removal of liquid from the condensation enclosure shall be collected from the drip pan and the total amount of such condensation collected throughout the test period shall be recorded in milliliters or fluid ounces. The exterior of the condensation enclosure shall not be touched, wiped, or otherwise disturbed during the replenishment of ice or removal of liquid. Condensation removed from the drip pan after ice replenishment or fluid removal shall not constitute failure of the test. The condensation enclosure shall be hung above the drip pan in a controlled temperature and humidity chamber so that one corner is hanging lower than the others. The controlled temperature and humidity chamber shall be capable of maintaining an air temperature of 14.4 to 16.7 °C (58 to 62 °F) and a relative humidity of 41 to 45 percent (see 4.5.3.4).

4.5.3.4 Humidity control. The chamber shall be capable of maintaining the required humidity throughout the course of the test. The controlled temperature and humidity chamber shall utilize an automated temperature humidity chamber, an aqueous solution in accordance with ASTM E104, or through a method approved by the Technical Authority. If the aqueous slurry solution in accordance with ASTM E104 is used to maintain chamber humidity, the salt shall be mixed with potable water to create a slurry in accordance with ASTM E104 and placed in a container in the chamber. The solution shall be maintained as a slurry for the duration of the test by adding salt or potable water to maintain the required test conditions (see 4.5.3.3). The time required to add salt or potable water to the chamber shall not exceed 15 minutes and any deviation from relative humidity requirements (see 4.5.3.3) during this period shall not constitute failure of the test.

4.5.3.5 Test criteria and replicates. During testing, the potable water-ice mixture or other liquid solution shall be replenished as necessary to maintain a temperature between 0 to 3.3 °C (32 to 38 °F) and shall be maintained at a volume of not less than half of the inner volume of the condensation enclosure. The condensation testing shall be performed a total of three times with either three different condensation enclosures or performed three times with the same enclosure with the coating removed and new coating applied in accordance with 4.5.3.2. The test setup shall be visually inspected (without magnification) for corrosion and coating failure (see 3.4.1) on the condensation enclosure and the drip pan shall be visually inspected for any condensation visible on the drip pan after the specified test time (i.e., 24 hours for grade A or 8 hours for grade B). Results shall conform to the requirements of 3.4.1.

4.5.4 Dry time. The coating shall be applied to achieve the manufacturer's maximum WFT (see 3.3.l) to a panel prepared in accordance with 4.5.1.1 under controlled laboratory conditions (see 4.5.1.2). The coating shall meet the dry-through requirement in accordance with ASTM D1640/D1640M, method A within 24 hours. Results shall conform to the requirements of 3.4.2.

4.5.5 Sag resistance. The coating shall be applied at a WFT of 60 mils. The sag resistance of the coating shall be determined in accordance with ASTM D4400 by applying the coating material to two panels prepared in accordance with 4.5.1.1 under controlled laboratory conditions (see 4.5.1.2). Results shall conform to the requirements of 3.4.3.

4.5.6 Knife adhesion. The coating shall be applied to achieve the manufacturer's maximum final system DFT (see 3.3.e) to three panels prepared in accordance with 4.5.1.1 under controlled laboratory conditions (see 4.5.1.2). Fully coated test specimens shall be aged 7 days under controlled laboratory conditions (see 4.5.1.2) before testing. The coating shall be tested for removal from the primer and shall be performed in accordance with ASTM D6677. The coating shall be applied at the manufacturer's maximum WFT per coat (see 3.3.l) over itself and aged for 7 days under controlled laboratory conditions (see 4.5.1.2) before testing. The coating shall be tested for removal from the previous coat in accordance with ASTM D6677. Results shall conform to the requirements of 3.4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.

4.5.7 Thermal conductivity. The thermal conductivity of class 1 and class 2 coatings shall be determined in accordance with ASTM C518 or through a method approved by the Technical Authority. Results shall conform to the requirements of 3.4.5.

4.5.8 Flash point. The flash point shall be measured in accordance with ASTM D3278, ASTM D6450, or through a method approved by the Technical Authority. Results shall conform to the requirements of 3.4.6.

4.5.9 VOC. The VOC for all types, classes, grades, and applications shall be determined in accordance with 40 CFR 60, appendix A-7, method 24. Results shall conform to the requirements of 3.4.7.

4.5.10 Mold/mildew resistance. The coating shall be applied to achieve the manufacturer's maximum final system DFT (see 3.3.e) to three panels prepared in accordance with 4.5.1.1 under controlled laboratory conditions (see 4.5.1.2). The test shall be performed in accordance with ASTM D3273. Results shall conform to the requirements of 3.4.8.

4.5.11 Fire performance. All fire tests specified in this document (see 4.5.11.1.1 through 4.5.11.1.3 for application A and 4.5.11.2.1 through 4.5.11.2.3 for application B) shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is acceptable to the Technical Authority. Accreditation shall be obtained from a recognized accreditation body such as the American Association for Laboratory Accreditation (A2LA) (www.a2la.org) or the International Code Council's International Accreditation Services (IAS) (www.iasonline.org). The scope of accreditation shall include specific flammability and fire tests (see 6.2 and 6.5).

4.5.11.1 Application A fire performance. The results shall conform to the requirements of 3.4.9.1.

4.5.11.1.1 Surface flammability. The coating shall be applied to achieve the manufacturer's maximum final system DFT (see 3.3.e). The test shall be performed in accordance with IMO Fire Test Procedures (FTP) Code, annex 1, part 5. Results shall conform to the requirements of 3.4.9.1.1.

4.5.11.1.2 Smoke density. The coating shall be applied to three panels to achieve the manufacturer's maximum final system DFT (see 3.3.e). If multiple coats are required to achieve the test thickness, the coats shall be applied in accordance with manufacturer's product data sheet. Fully coated test specimens shall be aged 14 days before testing. The test shall be conducted in accordance with ASTM E662 in both flaming and non-flaming modes for smoke density. Results shall conform to the requirements of 3.4.9.1.2.

4.5.11.1.3 Fire gas toxicity. The coating shall be applied to achieve the manufacturer's maximum final system DFT (see 3.3.e) in accordance with T9070-AK-DPC-010/078-1. If multiple coats are required to achieve the test thickness, the coats shall be applied in accordance with manufacturer's product data sheet. When tested in accordance with T9070-AK-DPC-010/078-1, the results shall conform to the requirements of 3.4.9.1.3.

4.5.11.2 Application B fire performance. Results shall conform to the requirements of 3.4.9.2.

4.5.11.2.1 Critical radiant flux. The coating shall be applied to a prepared test panel (see 4.5.1.1 and 4.5.1.2) to achieve the manufacturer's maximum final system DFT (see 3.3.e). If multiple coats are required to achieve the test thickness, the coats shall be applied in accordance with manufacturer's product data sheet. The test shall be conducted in accordance with IMO Fire Test Procedures (FTP) Code, annex 1, part 5. Results shall conform to the requirements of 3.4.9.2.1.

4.5.11.2.2 Smoke density. The coating shall be applied to a prepared test panel (see 4.5.1.1 and 4.5.1.2) to achieve the manufacturer's maximum final system DFT (see 3.3.e). If multiple coats are required to achieve the test thickness, the coats shall be applied in accordance with manufacturer's product data sheet. The test shall be conducted in accordance with ASTM E662 in both flaming and non-flaming modes for smoke density. Results shall conform to the requirements of 3.4.9.2.2.

4.5.11.2.3 Fire gas toxicity. The coating shall be applied to achieve the manufacturer's maximum final system DFT (see 3.3.e) in accordance with T9070-AK-DPC-010/078-1. If multiple coats are required to achieve the test thickness, the coats shall be applied in accordance with manufacturer's product data sheet. When tested in accordance with T9070-AK-DPC-010/078-1, the results shall conform to the requirements of 3.4.9.2.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.

4.6 Qualitative verifications.

4.6.1 Color. The coating shall be applied to achieve the manufacturer's maximum final system DFT (see 3.3.e) to one panel prepared in accordance with 4.5.1.1 under controlled laboratory conditions (see 4.5.1.2). The coating shall be visually inspected without magnification after 24 hours. The color shall visually match to the color identified and as defined by the manufacturer (see 3.3.a). Results shall conform to the requirements of 3.5.1.

4.6.2 Water resistance. The coating shall be applied to achieve the manufacturer's maximum final system DFT (see 3.3.e) to two panels prepared in accordance with 4.5.1.1 under controlled laboratory conditions (see 4.5.1.2) and allowed to dry for 24 hours. One panel shall be completely submerged in potable water for 1 hour. The panel shall be removed from the potable water and patted dry with a lint free cloth. The panel shall be visually inspected without magnification for blemishes, blisters, wrinkles, or delaminations. The color of the panel shall be compared to the non-submerged panel. Results shall conform to the requirements of 3.5.2.

4.6.3 Compatibility with primers and undercoats. The coating shall be applied to achieve the manufacturer's maximum final system DFT (see 3.3.e) to panels prepared in accordance with 4.5.1.1 under controlled laboratory conditions (see 4.5.1.2). The panels shall be visually inspected without magnification for blemishes, blisters, wrinkles, or delaminations. The coating shall be examined for incompatibility immediately after topcoat application and after 48 hours. The coating shall be applied again over itself and examined for blemishes, blisters, wrinkles, delaminations, or other signs of incompatibility immediately after topcoat application and after 48 hours. Results shall conform to the requirements of 3.5.3.

4.6.4 Accelerated storage stability. Two unopened, sealed containers of test coating shall be stored at 49 °C (120 °F) for 30 days. The sealed containers shall be removed from elevated temperature storage and shall be equilibrated to controlled laboratory conditions (see 4.5.1.2). The containers shall be opened, and the coating shall be tested in accordance with ASTM D1849 and FED-STD-141, method 3011.3, and the results shall conform to the requirements of 3.5.4.

4.6.5 Spraying properties. The coating shall be applied to two test panels that are prepared in accordance with 4.5.1.1 and 4.5.1.2. The coating shall be spray applied in accordance with FED-STD-141, method 4331.2. The dried coating shall be visually inspected without magnification. The dried coating shall be uniform and free of visual defects including dry spray. Results shall conform to the requirements of 3.5.5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2).

6. NOTES

INFORMATION FOR GUIDANCE ONLY. (This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. This specification covers anti-sweat coatings used to minimize or control condensation on a primed or painted interior surface. The coatings are not to be used as an alternative to conventional fiberglass or foam insulation. The anti-sweat coating is applied in thicker layers than other interior coatings. The coating does not contain asbestos and when dried is nonhazardous. The anti-sweat coatings are to be applied by spray, but brush application can be used to repair or touch up the coating. Existing stock is to be applied before additional anti-sweat coating is ordered.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, grade, and application of coating required (see 1.2).
c. Size of container required.
d. Color required.
e. Requirements for an HHA (see 3.1.2 and 4.5.2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.

f. Requirement for prohibited materials verification (see 3.1.3 and 4.5.2.3).

g. Requirement for HAPs content (see 3.1.3 and 4.5.2.4).

h. When first article is required (see 3.2, 4.3, and 6.3).

i. Requirement for ID characteristics (see 3.3).

j. Additional fire testing provisions (see 3.4.9, 4.5.11, and 6.5).

k. When quality conformance inspection is required (see 4.4).

l. Packaging requirements (see 5.1).

m. Requirements met by extension (see 6.7).

n. Required storage and transportation temperature range.

6.3 Inspection for first article. When a first article inspection is required, the items should be a first article sample. The contracting officer should include specific instructions (see 6.2) regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Toxicity evaluation. The NMCFHPC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCFHPC to the Government for evaluation. The HHA process is described on the NMCFHPC's website, https://www.med.navy.mil/Navy-and-Marine-Corps-Force-Health-Protection-Command/Environmental-Health/Industrial-Hygiene/Acquisition-Technical-Support/Health-Hazard-Assessments-HHA/.

6.5 Additional fire testing provisions. The Technical Authority reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements. A2LA (www.a2la.org) or International Code Council's IAS (www.iasonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

6.6 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.7 Meeting requirements by extension. Unless otherwise specified (see 6.2), coatings will meet the requirements of this specification by extension in the following manner:

a. Coatings that meet the VOC requirements for type I will meet the VOC requirements for type II and III by extension (see 3.4.7).

b. Coatings that meet the VOC requirements for type II will meet the VOC requirements for type III by extension (see 3.4.7).

c. Coatings that meet the thermal conductivity requirements for class 1 will meet the thermal conductivity requirements for class 2 by extension (see 3.4.5).

d. Coatings that meet the condensation requirements for grade A will meet the condensation requirements for grade B by extension (see 3.4.1).

6.8 Coating thickness. The use thickness of the coating passing the requirements of this specification will be set by the manufacturers [mDFT (see 3.3.m)] as required to meet the condensation test (see 3.4.1). The maximum use thickness of the coating passing the requirements of this specification will be set by the fire performance testing (see 3.4.9).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.

6.9 Subject term (key word) listing.

Anti-condensation
Condensation
Corrosion
Fire performance
Insulation
Paint
Sweat

6.10 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

CONCLUDING MATERIAL

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
Navy – SH
(Project 8010-2020-011)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:16Z
Check the source to verify that this is the current version before use.